PANJABUNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd.underthePanjabUniversityActVIIof1947 –enacted by the Govt. of India)

SYLLABI

Course Outline for

B.Voc. (Horticulture Sciences)

As per NEP 2020

W.e.f. session 2024-25

Course Outline for B.Voc. (Horticulture Sciences)
As per NEP 2020 w.e.f. session 2024-25

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HSC 101	Core Discipline	Weed Control	2	2	4	40	10	40	10	100
HSC 102	Core Discipline	Nursery Production	2	2	4	40	10	40	10	100
HSC 103	Core Discipline	Irrigation and Water Management	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 101	Multidisciplinary Course	Fundamentals of Information Technology[1]	3	-	3	60	15	-	-	75
SEC I	Skill Enhancement Course	SEC Options[2]			3					75
VAC I	Value-Added Course	VAC Options[3]			2					50
PW I	Project Work	Project Work-I& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
HSC 201	Core Discipline	Propagation Techniques	2	2	4	40	10	40	10	100
HSC 202	Core Discipline	Elements of Vegetable Growing	2	2	4	40	10	40	10	100
HSC 203	Core Discipline	Horticulture (Fruit Growing)	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 201	Multidisciplinary Course	Soft Skills and Personality Development[1]	3	-	3	60	15	-	-	75
SEC II	Skill Enhancement Course	SEC Options[2]			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role - Gardener and Nurseryman

*Summer Industrial Training of 2-4 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of the start of the 3rd Semester. Viva-Voce examination to beheld within 3 weeks of the start of 3rd semester.

** Minimum Credits required to get Certificate Course in Horticulture Sciences = 48 + 4 #.

#Exit 1: Certificate Students can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during the first and second semester.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

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ESSENTIAL COMPONENTS OF COURSE OF B.VOC

B.Voc. (Horticulture Sciences)

(SEMESTERS I &II)

PROGRAMME OUTCOMES

PO1. To transfer knowledge of Agriculture/Horticulture in the field of agricultural research, especially in Horticulture including Fruits, Vegetables, Flowers, Species, Medicinal & Aromatic Plants & their management.

PO2. To develop innovative agro-techniques to enhance the production & productivity of Horticulture Crops.

PO3. To increase farmer's income through Hi-tech Horticulture.

PO4.To create job opportunities for unemployed youth through teaching, research,training, extension, etc.

PO5. To the establishment of model nurseries in rural areas for the availability of qualityplantingmaterials.

PO6. To prolong the post-harvest storage life of Horticulture commodities & increase incomethrough value addition of the products & to reduce post-harvest losses.

PO7. To develop interpersonal skills under the students to identify, analyze & solve problems.

Semester I

	COURSE TITLE	WEED CONTROL
1.	**Course Code**	HSC 101
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lecture) (Practical)**	30 hours of lectures for Theory + 60 hours of practical(1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by theUniversity/College to the examinees
6.	**Job Role**	Gardener & Nursery Man.
7.	**Course Objectives**	To acquaint the students about elementary information on basic agronomic principles and practices for horticulture crop.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. To understand about harmful & beneficial effects of weed in horticulture. CO2. To identify different weeds, cycle of weeds & crop competition. CO3. To understand different methods of weed control and Integrated Weed Management. CO4. To identify and apply herbicides for the control of weeds in different horticulture crops CO5. To understand the approaches of weed management in different cropping systems
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical andGroup Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT– I

Definition of weeds, their characteristics & harmful effects. Classification of weeds, propagation & dissemination in weeds.

UNIT– II

Herbicide structure and properties, factors affecting the efficiency of herbicides, herbicide formulations, herbicide mixtures, Degradation of herbicides in soil and plants, weed shift in cropping systems, weed control through bioherbicides and allelochemicals, herbicide resistant crops,herbicide rotation.

UNIT– III

Weed control/ management (Legislative, physical, chemical, mechanical, biological) in cropping systems and non – cropped situations —control of parasitic weeds – control of aquatic, biennial and perennial weeds.

UNIT-IV

Weed control/ management in major horticultural crops. Integrated weed management,

12. Suggested Resources for Reading

1. Ashton, F.M and Crafts, A.S. 1981. Mode of Action of Herbicides (2nd Ed). Wiley Inter-Science
2. Gupta, O.P. 2007. Weed Management – Principles and Practices. Agro bios
3. Mandal, R. C, 1990. Weed, Weedicides and Weed Control – Principles and Practices. Agro–Botanical Publ.
4. Rao, V.S. 2000. Principles of Weed Science. Oxford & IBH

Note: Latest edition of textbooks may be used.

	COURSE TITLE	NURSERY PRODUCTION
1.	**Course Code**	HSC 102
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week)**(Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) **(**1 Semester = 15 weeks) 3-groups will be created for Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (80TH/PR+20IA) i) The exam of this course will be of 3 hours duration. ii) There will be no objective-type questions. iii) Students are required to know about the developments in thesubject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Self Employment in Business of Nursery.
7.	**Course Objectives**	To make the students familiar with the basics principles and procedures of nursery rising for horticulture crops.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Familiarization with principles & practices of propagation & nursery management for Horticultural Crops. CO2. Study of introduction to propagation, cellular basis for propagation, sexual propagation, apomixes, polyembryony, chimeras. CO3. Principal factors influencing seed germination of horticulture crops, dormancy, hormonal regulation of germination & seedling growth. CO4. Knowledge of nursery management, nursery establishment & nursery rules & regulations. CO5. Imparting knowledge about varieties and F1 hybrids – seed rate – seed treatments –sowing – media management for water, nutrients and plant protection, hardening, packaging, storage, transport, and marketing of vegetables.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies, etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) The examiner will set a total of NINE questions comprising two questions each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and theentire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT– I

Scope and importance of vegetable and flower nursery planning I - location, soil, topography, layout, paths, roads, nursery beds, office, store, potting yard planning II - water, containers, electricity, fencing, shelter belts – Propagation structures - greenhouses, Mist chambers, poly houses – growing media - selection of media, quality of media, soil, sand, peat, leaf mold, saw dust, sphagnum moss, coco peat.

UNIT– II

Propagation – seed germination – dormancy-methods of breaking dormancy – pre-sowing seed treatment. methods of sowing – irrigation systems- nutrient management – water soluble and liquid fertilizers

UNIT– III

Varieties and F1 hybrids – seed rate – seed treatments –sowing – media management for water, nutrients, packaging, storage, transport, and marketing of tomato, brinjal, chili, capsicum, cabbage, cauliflower, carrot, radish.

UNIT-IV

Varieties and F1 hybrids – seed rate – seed treatments –sowing – media management for water, nutrients, packaging, storage, transport, and marketing of mango, apple, pear, peach, and guava.

12. Suggested Resources for Reading

1. Nanda, K.K. and V.K. Kochhar; 1991. Vegetative propagation and management. Indus Publishing Company, New Delhi.
2. Sadhu, N.K. 1989. Plant propagation, Wiley Eastern Ltd., New Delhi.
3. Agarwal, R.L. 2005. Seed Technology. Oxford and IBH Publishing Co. New Delhi.

Note: Latest edition of textbooks may be used.

	COURSE TITLE	IRRIGATION AND WATER MANAGEMENT
1.	**Course Code**	HSC 103
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practicals for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (80TH/PR+20IA) i) The exam of this course will be of 3 hours duration. ii) There will be no objective-type questions. iii) Students are required to know about the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by theUniversity/College to the examinees
6.	**Job Role**	Technician in Irrigation Department.
7.	**Course Objectives**	To acquaint students with a soil-water relationship in the context to agriculture/ horticulture crops.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Impart knowledge and understanding of fundamental concepts and techniques of maintaining and enhancing water management. CO2. Develop the ability to skill the practices and their scope to improve inthe water management. CO3.Impart knowledge and understanding of sources, modes, and systems of irrigation CO4. Familiarization with different methods of water conservation**.**
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, Practical and Group Discussions, Case Studies, etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and theentire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT– I

Irrigation and its importance, Importance of water to plants, Time of irrigation. Principal of irrigation (soil moisture, root zone depth, water holding parameters, effective irrigation, daily water use) Basis for scheduling irrigation to crops: plant basis approach, climatologically approach and soil moisture depletion approach.

UNIT– II

Sources and modes of irrigation (canals, wells, tube wells, tank irrigation), irrigation system of Punjab, Flood irrigation, furrow irrigation, sub surface irrigation, overhead irrigation, manual irrigation, localized irrigation and sprinkler irrigation. Methods of measurement of irrigation, water requirement of different crops.

UNIT– III

Water logging, different types of water movement in soil. Principal of drainage: surface drainage and sub surface drainage. Kind of sub surface drains. Crop response to quality of irrigation water. Irrigation management and problem, Soil irrigation for frost protection.

UNIT-IV

Erosion Problems in Punjab and India, type of Erosion: water Erosion, Sheet and Rill Erosion, Scalding Erosion, Gully erosion, Tunnel Erosion, Stream Bank Erosion, Mass Erosion and Wind Erosion, Causes and Effects of Erosion, Factors responsible for Water and Wind Erosion. Form of wind and water Erosion. Different methods of water conservation.

12. Suggested Resources for Reading

1. Iraeslon, Irrigation Principles, John Willey & Sons New Delhi 1996.
2. Gandhi R T., Gupta, P.C., Joseph, A.P. and Rage, N. I, Handbook of Irrigation WaterManagement.
3. Mickael, A.M., Irrigation Theory and Practices.

Note: Latest edition of text books may be used.

Semester II

	COURSE TITLE	PROPAGATION TECHNIQUES
1.	**Course Code**	HSC-201
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week)**(Practical**)	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100 Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed.The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Planter.
7.	**Course Objectives**	To get basic knowledge of various propagation techniques to develop practical skills.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. Familiarization with principals & practices of propagation & nursery management for Horticultural Crops. CO2. Study of introduction to propagation, cellular basis for propagation,sexual propagation, apomixes, polyembryony, chimeras. CO3. Principal factors influencing seed germination of horticulture crops, dormancy, hormonal regulation of germination & seedling growth.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidate**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus**.** iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT– I

Propagation: Need and Potentialities, Sexual and Asexual methods of propagation and their advantages and disadvantages Seed Dormancy: Causes (Internal and external factors) Seed Dormancy: Methods of breaking Dormancy

UNIT– II

Cutting Techniques: Stem Cutting, Leaf Cutting, root cutting Layering Techniques: tip layering, simple layering, compound layering, mound layering, air layering

UNIT– III

Grafting techniques, necessity of Grafting in plants and backstop. Type of Grafting: cleft Grafting, Bark Grafting, Side-veneer Grafting, Splice Grafting, Whip and Tongue Grafting, Saddle Grafting, Bridge Grafting.

UNIT-IV

Methods of Vegetative Propagation: Budding techniques: T or Shield Budding, Chip Budding, Inverted T-Budding, I-Budding, Flute Budding, Patch Budding. Advantages of Budding and safeguard during budding.

12. Suggested Resources for Reading

1. Bose, T.K. Sanyal, D and Sandhu, M.L. (1998) Propagation of Horticultural crops. Naya Prakash Publishers, Kolkata
2. Sadhu, N.K. 1989. Plant propagation, Wiley Eastern Ltd., New Delhi.
3. Nanda K. K., Kochhar V.K. (1985). Vegetative Propagation of Plants. Kalyani Publisher- New Delhi-Ludhiana.
4. Hartmann, H.T., Kester D.E., Davis, F.T and R.L Geneve (2010). Plant Propagation: Principles and Practices (8th Edition)
5. Grewal, H.S. 1999. Propagation of ornamental plants, Kalyani Publishers, New Delhi
6. Hartmann, H.T. and D.E. Kester; 1989. Plant propagation-Principles and practices. Prentice – Hallof India, New Delhi.
7. Nanda, K.K. and V.K. Kochhar; 1991. Vegetative propagation and management. Indus Publishing Company, New Delhi.

Note: Latest edition of text books may be used.

	COURSE TITLE	ELEMENTS OF VEGETABLE GROWING
1.	**Course Code**	HSC-202
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100 Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
6.	**Job Role**	Supervisor in Vegetable Industry.
7.	**Course Objectives**	To get cultivations and other agronomic practices for raising vegetable crops.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. To educate production technology of seasonal vegetables. CO2. To imparting knowledge about botany & taxonomy, climate & soil requirements, commercial varieties/hybrids, sowing/planting time & methods, seed rate & seed treatment, nutritional & irrigation requirements, intercultural operations, weed control, mulching, physiological disorders, harvesting, post harvest management, plant protection measures, & seed production of vegetables.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and theentire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT– I

Scope, Importance and Types of vegetable growing, importance of kitchen garden. Classification of vegetablecrops.

UNIT– II

Cultivation practices including Soil and Climatic requirements, varieties, seed rate, time of sowing, spacing, fertilizers and manures, irrigation, hoeing and yield potential of Cole Crops, Root Crops, Bulb Crops and theirharvesting.

UNIT– III

Cultivation practices including Soil and Climatic requirements varieties, seed rate, time of sowing, spacing,fertilizers and manures, irrigation, hoeing, and yield potential of Peas, Okra, eggplant and their harvesting.

UNIT-IV

Storage and seed production of the Potato, Cole Crops, Root Crops, Bulb Crops Peas and Solanaceous fruits.

12. Suggested Resources for Reading

1. Dhaliwal M.S., Handbook of vegetable crops, Kalyani Publishers, Ludhiana, 2008.
2. Package of Practices for Vegetable crops, P.A.U. Publications Ludhiana, Corresponding year.
3. Das, P.C., Vegetable crops of India, Kalyani Publishers, Ludhiana, 1993.
4. Chauhan, D.V., Vegetable production in India, S. Ram Prasad and Sons, Agra, 1993

Note: Latest edition of textbooks may be used.

	COURSE TITLE	HORTICULTURE (FRUIT GROWING)
1.	**Course Code**	HSC-203
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures)** hrs./week)**(Practical)**	30 hours of lectures for Theory + 60 hours of practicals (1 credit = 15 hours of theory learning based on 2 (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks**)** 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100 Marks (80TH/PR+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective-type questions. iii) Students are required to have knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by theUniversity/College to the examinees
6.	**Job Role**	Horticulture Laboratories & Field Technician.
7.	**Course Objectives**	To learn agronomic and other management practices in fruit growing. The studentwill also learn fruit preservation method & their scope.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. To understand the principle, and theoretical aspects & developing skills infruit cultivation or horticultural crops. CO2.To impart basic knowledge about the importance & management of fruits grown in India. CO3. Study of commercial varieties of regional, national & international importance, eco-physiological requirements, recent trends in propagation, rootstock influence, planting system, cropping systems, cropping system, root zone & canopy management, nutrient management, water management, fruit set & development, etc. CO4. To execute quality improvement by management practices.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and theentire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT– I

Fruits and their importance, Selection of site and soils for planting an orchard. Orchard soil management: clear culture, mulching, sod, sod mulch, intercropping, mixed cropping, multitier cropping, organic farming. Growing ofintercrops as green manure and cover crops, the mulching effect of soil moisture.

UNIT– II

Importance and scope of fruit growing in Punjab such as increase production, increase irrigation facilities, increase area under dry land, increase urbanization, increase transport facilities, increase cold storage facilities and pre cooling center, development new techniques, availability of loan, government incentive, export of fruits.

UNIT– III

History and development of fruit preservation and its scope. Canning of fruit (Gooseberry, Blueberry, Blackberry, Apple, Pear) Bottling of fruits (Banana jam, Lemon curd, Spiced citrus fruit, pickled Pears, Blueberry jam). Cold storage of fruits and types: blast freezer and chiller, cold room, refrigerated and transportation method.

UNIT-IV

Different methods (canning, cooling, pasteurization, heating, boiling and adding sugar, salt and additives) of preservation and preparing jams, jellies and marmalades. Role of plant growth regulators in fruit production: Abscisic acid, Auxin, Cytokinins, Ethylene and Gibberellins.

12. Suggested Resources for Reading

1. Singh, R., Fruits, National Book Trust of India, New Delhi, 1969.
2. Package of Practices for Cultivation of Fruits, P.A.U. Publications Ludhiana,Corresponding year.
3. Hayes, W.B., Fruit growing in India, Kitabistan, Allahabad, 1945.
4. Lal, G., Siddappa, S. And Tandon, G.L., Preservation of fruits and vegetables, Indian Council of Agricultural Research, New Delhi, 2009.

Note: Latest edition of text books may be used.